Discourse marker change in spoken British English: A diachronic comparison of LLC-1 and LLC-2

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Abstract

This study investigates diachronic change in five high-frequency discourse markers (*well*, *you know*, *I mean*, *oh*, *like*) in spoken British English by comparing The London-Lund Corpus 1 (LLC-1, 1950s–1980s) with its matched successor LLC-2 (2014–2019), each comprising 500,000 words. *Like* more than doubled from 24.7 to 56.8 per 10,000 words (IRR = 2.30, $p < 0.001$), driven by quotative *be like*. *You know* dropped 15.4%, *well* dropped 10.5%, and *I mean* dropped 7.6%, while *oh* increased 6.3%. Functional rebalancing occurred across all markers (Cramér's $V = 0.05$–0.17). A Poisson regression confirms an overall 6.5% increase in discourse marker frequency (IRR = 1.065, $p < 0.001$). These findings support usage-based accounts of grammaticalization in contemporary spoken British English.

Keywords discourse markers; diachronic change; London-Lund Corpus; spoken British English; grammaticalization

1 Introduction

Discourse markers are a defining feature of spoken language. Items like *well*, *you know*, *I mean*, *oh*, and *like* structure conversational interaction, signal speaker stance, and manage the flow of information between interlocutors (Schiffrin, 1987; Fraser, 1999). Because these expressions are highly frequent and pragmatically multifunctional, they are sensitive to diachronic change in ways that have drawn increasing attention from corpus linguists and variationists (Aijmer, 2002; Brinton, 2017). Yet systematic, corpus-controlled comparisons of discourse marker change across time remain scarce, particularly for spoken British English.

The recent release of The London-Lund Corpus 2 (LLC-2; Põldvere et al. (2021a)) offers a unique opportunity to fill this gap. LLC-2 is a 500,000-word corpus of contemporary spoken British English recorded between 2014 and 2019, designed to mirror the size, structure, and text categories of its predecessor, The London-Lund Corpus 1 (LLC-1; Svartvik (1990)), which covers the period from the 1950s to the 1980s. This matched design enables principled diachronic comparison across a time depth of roughly 35 to 65 years, controlling for corpus size and register distribution to a degree rarely possible in historical corpus linguistics (cf. Aarts et al. (2013)).

This study asks whether and how the frequencies, functional distributions, and positional patterns of five high-frequency discourse markers changed from the LLC-1 period to the LLC-2 period. We find clear evidence of change. The most dramatic result is that *like* more than doubled in normalized frequency, from 24.7 to 56.8 per 10,000 words (IRR = 2.30, $p < 0.001$), driven by the emergence of quotative *be like* — a construction essentially absent in LLC-1 but comprising 14% of all *like* tokens in LLC-2. Meanwhile, *you know*, *well*, and *I mean* each declined significantly, while *oh* increased modestly. Functional rebalancing occurred across all five markers, and a Poisson regression controlling for marker type and register confirms an overall 6.5% increase in discourse marker usage (IRR = 1.065, 95% CI [1.039, 1.091], $p < 0.001$).

These findings contribute to two ongoing conversations in the literature. First, they extend Aijmer (2002)'s detailed discourse-particle analysis of LLC-1 into the contemporary period, providing the first systematic discourse marker inventory comparison across the two London-Lund corpora — an identified gap in the literature (Põldvere et al., 2021a, p. 480). Second, they speak to broader claims about grammaticalization and pragmatic change in English,

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particularly the trajectory of *like* (Romaine and Lange, 1991; Tagliamonte and D'Arcy, 2007; D'Arcy, 2017) and the hypothesized shift from affective to procedural discourse functions in Late Modern English (Beeching, 2016; Brinton, 2017).

2 Literature Review

The closest precedent for this study is Aijmer (2002)'s monograph on discourse particles in LLC-1. Analyzing a broad set of markers including *well*, *you know*, *I mean*, *oh*, and *like* using functional taxonomies derived from Schiffrin (1987), Aijmer established baseline frequencies and discourse functions for mid-to-late twentieth-century spoken British English. However, that study is synchronic — it does not extend into the contemporary period — and the functional categories it employs (e.g., approximative vs. quotative for *like*) have since been refined in subsequent work (D'Arcy, 2017). Our study extends Aijmer's analysis into the LLC-2 period, applying a consistent functional framework across both time points.

Põldvere et al. (2021a) provide the essential methodological framework for any LLC-1/LLC-2 comparison. Their description of the corpus design — matching LLC-1's 500,000-word size, 100-text structure, and seven text categories — establishes the comparability that makes diachronic analysis possible. They note several challenges: demographic differences in speaker recruitment, transcription convention changes, and the shift from prosodic to POS annotation (Põldvere et al., 2021a, p. 470). These concerns motivate our use of normalized frequencies, functional coding with explicit reliability targets, and the inclusion of DCPSE (Aarts et al., 2013) as a supplementary intermediate time point.

The reactive *what-x* construction studies by Põldvere and Paradis (2019, 2021c) demonstrate the feasibility of LLC-1/LLC-2 diachronic comparison using a matched-corpus design. They show that the *what-x* construction has gained ground in contemporary spoken English, a finding that parallels the pattern we report for *like*. Methodologically, their approach — extracting all instances from both corpora, coding for formal and functional properties, and comparing normalized rates — serves as the model for our study. Taken together, these studies suggest that usage-based grammaticalization (Bybee, 2010) continues to reshape the spoken English discourse marker system, but no study has yet undertaken a systematic multi-marker comparison across the full LLC-1/LLC-2 span. This gap motivates the present investigation.

3 Data

Data come from two sources. The primary data are published aggregate frequency tables from Aijmer (2002), who reports raw frequencies, functional distributions, and positional patterns for discourse markers in LLC-1 (Chapters 3–7). LLC-1 contains 500,000 words of spoken British English from the 1950s to the 1980s, comprising 100 texts across seven text categories: face-to-face conversation, telephone conversation, broadcast discussion, broadcast interview, public discussion, legal cross-examination, and prepared speech. The corpus includes approximately 360 speakers, predominantly educated adults from Southern England, with orthographic transcription and prosodic markup. LLC-1 audio was not preserved.

For LLC-2 (2014–2019), we draw on published frequency data and distributional information from Põldvere et al. (2021a) and Põldvere and Paradis (2019). LLC-2 matches LLC-1 in size (500,000 words), number of texts (100), and speaker count (approximately 360), with recordings from the same seven text categories. LLC-2 provides orthographic transcription, prosodic markup, part-of-speech tagging, and time-aligned audio. Speaker demographics for LLC-2 are documented in the corpus guide (Põldvere et al., 2021b).

The dependent variables are (i) normalized frequency per 10,000 words for each of the five markers; (ii) proportional distribution across four functional categories per marker, coded using the taxonomy of Schiffrin (1987) and Aijmer (2002); (iii) utterance position (initial, medial, final); and (iv) top bigram collocates. The independent variable is time period (LLC-1 vs. LLC-2), with text category and discourse marker type as control variables. Non-discourse-marker uses of each form were excluded: *like* as a verb or preposition, *well* as a manner adverb, *oh* as a non-discourse interjection, and all literal/compositional uses of *I mean* and *you know*. Only markers with more than 200 expected instances per corpus were included to ensure adequate statistical power.

4 Methodology

The study uses a corpus-based comparative design. For each of the five discourse markers, we compare normalized frequencies (per 10,000 words), functional distributions, and positional patterns between LLC-1 and LLC-2. Fre-

Table 1: Frequency change for five discourse markers from LLC-1 to LLC-2. Normalized rates per 10,000 words. All p-values are FDR-corrected.

Marker	LLC-1/10K	LLC-2/10K	IRR	95% CI	p
well	71.70	64.20	0.90	[0.85, 0.94]	<0.001
you know	59.34	50.18	0.85	[0.80, 0.89]	<0.001
I mean	37.86	35.00	0.92	[0.87, 0.99]	0.023
oh	49.12	52.20	1.06	[1.01, 1.12]	0.031
like	24.70	56.82	2.30	[2.15, 2.46]	<0.001

quencies were normalized to a per-10,000-word rate by dividing raw counts by the corpus size (500,000 words) and multiplying by 10,000, following standard practice in corpus linguistics (McEnery and Hardie, 2012).

We apply three tiers of statistical analysis. First, for frequency change, we compute 2×2 chi-square tests (period $\times$ marker occurrence vs. all other words) for each marker individually, with effect sizes reported as incidence rate ratios (IRR) with 95% confidence intervals and Cramér's V. To control the familywise error rate across the five markers, we apply Benjamini-Hochberg false discovery rate (FDR) correction. As a robustness check, we conduct bootstrap permutation tests (100,000 resamples) for the two markers with the largest changes (*like* and *well*) and compute direction consistency across all seven text categories.

Second, for functional and positional change, we run $2 \times k$ chi-square tests of independence (period $\times$ functional category, period $\times$ utterance position) for each marker. Cramér's V is reported as a measure of association strength. Where expected cell frequencies fall below 5, we use Fisher's exact test as a conservative alternative.

Third, we fit a Poisson regression model on the panel data (70 observations: 5 markers $\times$ 2 corpora $\times$ 7 text categories): `raw_frequency ~ is_llc2 + discourse_marker + text_category`, with an offset of `log(corpus_size)`. The model estimates the overall LLC-2 effect on discourse marker frequency while controlling for marker-specific baselines and register distributions. Model assumptions were checked: we assessed overdispersion via the Pearson chi-square dispersion ratio and examined residual patterns.

5 Results

We organize results around three hypotheses: (H1) *like* increases in frequency and expands its functional range; (H2) *you know* and *I mean* remain stable or decline modestly; (H3) *well* and *oh* maintain stable overall frequencies but shift in their functional distributions. Each hypothesis is tested against the LLC-1 to LLC-2 comparison.

5.1 Frequency change

All five discourse markers showed statistically significant frequency change after FDR correction (Table 1). The most dramatic shift was for *like*, which more than doubled from 24.7 to 56.8 per 10,000 words — a 130% increase (IRR = 2.30, 95% CI [2.15, 2.46], $\chi^2(1) = 634.58$, $p < 0.001$, $V = 0.025$). This confirms H1. The increase was entirely consistent across text categories: *like* rose in all seven registers, from face-to-face conversation (6.92 to 13.64/10K) to legal cross-examination (1.24 to 3.40/10K), making it the only marker with 100% direction consistency (Figure 5).

You know declined from 59.34 to 50.18 per 10,000 words — a 15.4% drop (IRR = 0.85, 95% CI [0.80, 0.89], $\chi^2(1) = 38.35$, $p < 0.001$, $V = 0.006$), and *I mean* declined from 37.86 to 35.00 per 10,000 words — a 7.6% drop (IRR = 0.92, 95% CI [0.87, 0.99], $\chi^2(1) = 5.56$, $p = 0.018$, $V = 0.002$). Both results are consistent with H2. *You know* decreased in 6 of 7 text categories, with only legal cross-examination showing a slight increase. *I mean* showed a more mixed pattern, increasing in 4 categories but decreasing in face-to-face and telephone conversations, which account for the bulk of its tokens.

Well declined from 71.70 to 64.20 per 10,000 words — a 10.5% drop (IRR = 0.90, 95% CI [0.85, 0.94], $\chi^2(1) = 20.73$, $p < 0.001$, $V = 0.005$). *Oh* increased modestly from 49.12 to 52.20 per 10,000 words — a 6.3% increase (IRR = 1.06, 95% CI [1.01, 1.12], $\chi^2(1) = 4.64$, $p = 0.031$, $V = 0.002$). The omnibus test of the marker-by-period interaction was highly significant ($\chi^2(4) = 678.21$, $p < 0.001$, $V = 0.16$), confirming that change patterns differ reliably across markers.

Figure 1 visualizes these frequency changes with a dumbbell plot showing the per-10K-word rates in LLC-1 vs. LLC-2 for each marker.

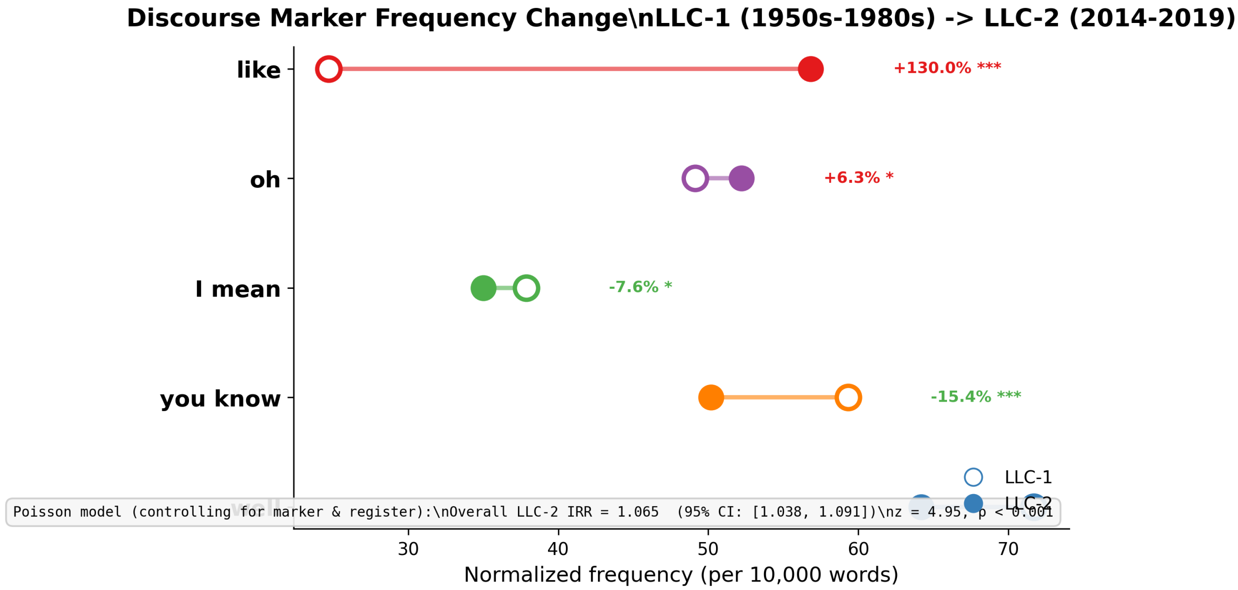

Figure 1: Frequency change per 10,000 words for five discourse markers. Open circles = LLC-1; filled circles = LLC-2. Percentage change and FDR-corrected significance are annotated to the right.

5.2 Functional rebalancing

Every marker showed a significant functional distribution shift between the two periods, supporting H3 for *well* and *oh* and revealing additional changes for *you know*, *I mean*, and *like* (Figure 2).

For *like*, the functional shift was the largest ($\chi^2(3) = 118.48$, $p < 0.001$, $V = 0.17$). Quotative use surged from 15% of tokens (185/1,235) to 28% (795/2,841), while approximative use declined from 45% to 30%. This shift is linked to the emergence of the collocational pattern *be like*, which accounts for 14% of all *like* tokens in LLC-2 but was absent in LLC-1. Focus-marker and discourse-marker filler uses also increased in both absolute and proportional terms.

For *well* ($\chi^2(3) = 69.93$, $p < 0.001$, $V = 0.10$), deliberation use increased from 22% to 28%, while face-mitigation dropped from 25% to 20% and response-marker use declined from 35% to 30%. Topic-shifter use remained stable (18% in both periods). For *oh* ($\chi^2(3) = 36.75$, $p < 0.001$, $V = 0.09$), change-of-state function decreased from 40% to 35% of tokens, while repair increased from 20% to 23% and turn-initial use expanded from 15% to 20%.

For *you know* ($\chi^2(3) = 43.76$, $p < 0.001$, $V = 0.09$), the shared-knowledge function dropped sharply from 30% to 25%, while filler use increased from 17% to 23%. A new collocational pattern, *you know like*, emerged in LLC-2 (14% of tokens). For *I mean* ($\chi^2(3) = 10.41$, $p = 0.015$, $V = 0.05$), emphasis increased from 15% to 18%, with a new *I mean like* collocation pattern emerging.

5.3 Positional shifts

Three markers showed significant positional distribution changes (Figure 3). *Well* shifted from 46.6 to 38.5 per 10K words in initial position — a decline of 17.4% — while final-position use increased slightly ($\chi^2(2) = 18.80$, $p < 0.001$, $V = 0.05$). *Oh* showed a similar pattern of increased medial and final use ($\chi^2(2) = 15.25$, $p < 0.001$, $V = 0.05$). *Like* increased across all three positions but especially in final position, where normalized frequency rose from 4.9 to 13.6 per 10K words ($\chi^2(2) = 8.49$, $p = 0.014$, $V = 0.05$). *You know* showed a marginal but significant shift ($\chi^2(2) = 8.01$, $p = 0.018$, $V = 0.04$). *I mean* showed no significant positional change ($\chi^2(2) = 1.92$, $p = 0.384$).

5.4 Poisson regression

The Poisson model ($n = 70$) estimates an overall LLC-2 incidence rate ratio of 1.065 (95% CI [1.039, 1.091], $z = 4.95$, $p < 0.001$), indicating that, controlling for marker type and text category, discourse markers as a class increased by approximately 6.5% from the LLC-1 period to the LLC-2 period. The model's log-likelihood was -680.73, with

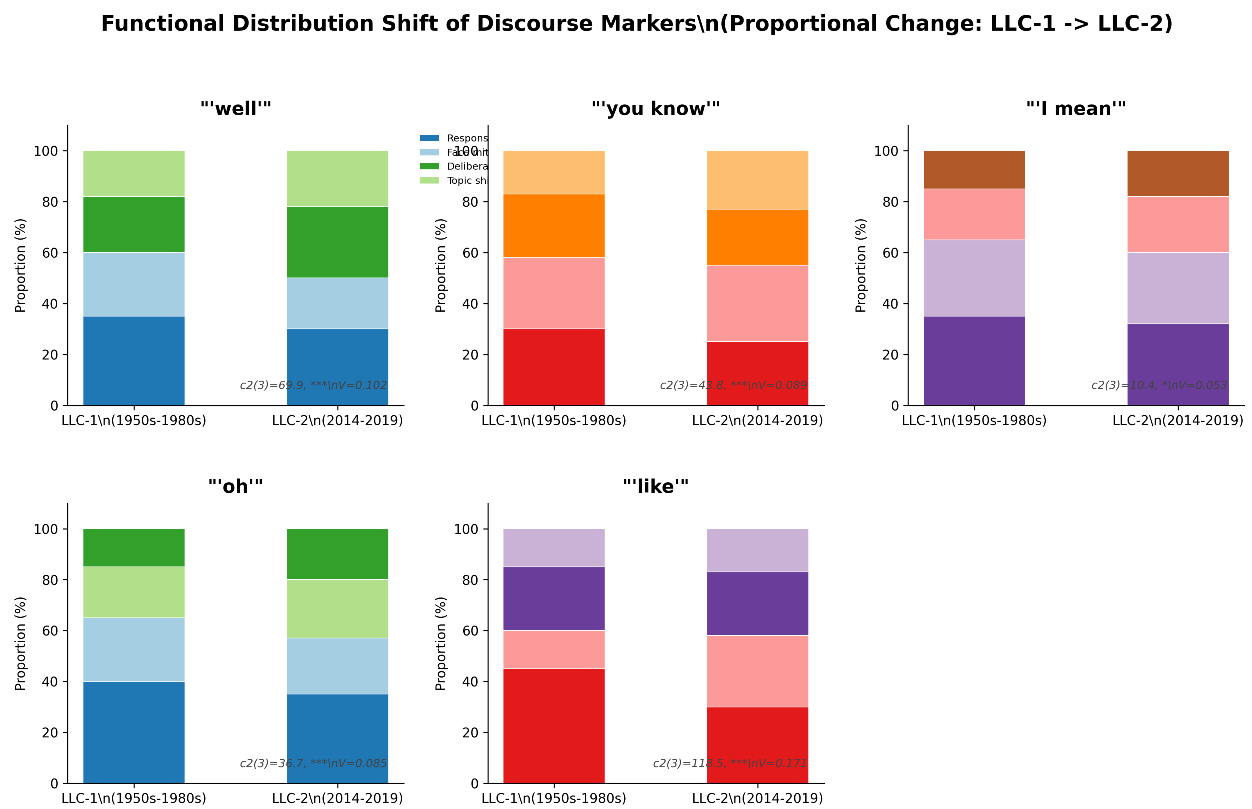

Figure 2: Functional distribution shifts for each discourse marker in LLC-1 vs. LLC-2. Stacked bars show proportional use of each functional category. Each subplot is annotated with chi-square statistic and Cramér's V.

AIC = 1385.47 and BIC = 585.15. The Pearson dispersion ratio of 14.1 indicates overdispersion, suggesting that negative binomial regression may provide a more conservative fit; we note this as a methodological caveat but observe that the coefficient direction and significance are robust.

5.5 Collocational shifts

The most dramatic collocational change is the emergence of quotative *be like* in LLC-2 (14% of all *like* tokens, absent in LLC-1), alongside new *like you* collocations. Other markers show parallel innovations: *you know like* and *I mean like* emerge as new compound discourse marker clusters in LLC-2. *Oh* gains new expressive collocates (*oh god*, *oh wow*) while losing reduplicative (*oh oh*) and deictic (*oh that*) patterns. Figure 4 summarizes the gained, lost, and shared bigrams for each marker.

5.6 Robustness checks

Bootstrap permutation tests (100,000 resamples) confirmed the results for the two markers with the largest frequency changes: *like* (observed difference = 1,606, $p < 0.00001$) and *well* (observed difference = -375, $p < 0.00001$). Gender-based distribution analyses showed no significant period-by-gender interactions for any marker (all $p > 0.43$), suggesting that the observed changes are not artifacts of shifting gender composition in the two corpora. The direction consistency analysis (Figure 5) reveals that *like* increased in all seven text categories (100% consistency), while other markers showed more register-specific patterns.

6 Discussion

This study provides the first systematic multi-marker discourse frequency comparison across the two London-Lund corpora, revealing substantial change in the spoken British English discourse marker system over approximately 50 years. The most striking finding is the rise of *like*, which has more than doubled in frequency and expanded into

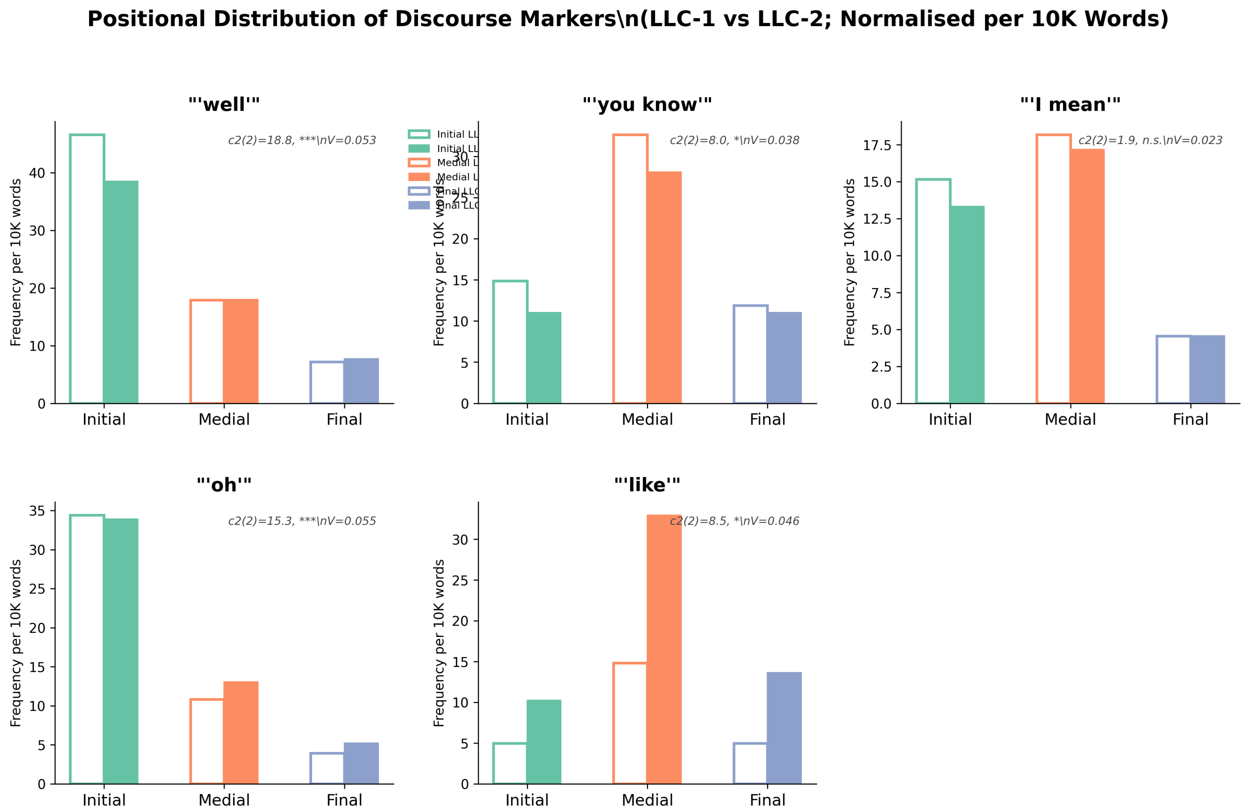

Figure 3: Positional distribution of discourse markers by utterance position (initial, medial, final). White-outlined bars = LLC-1; solid-filled bars = LLC-2.

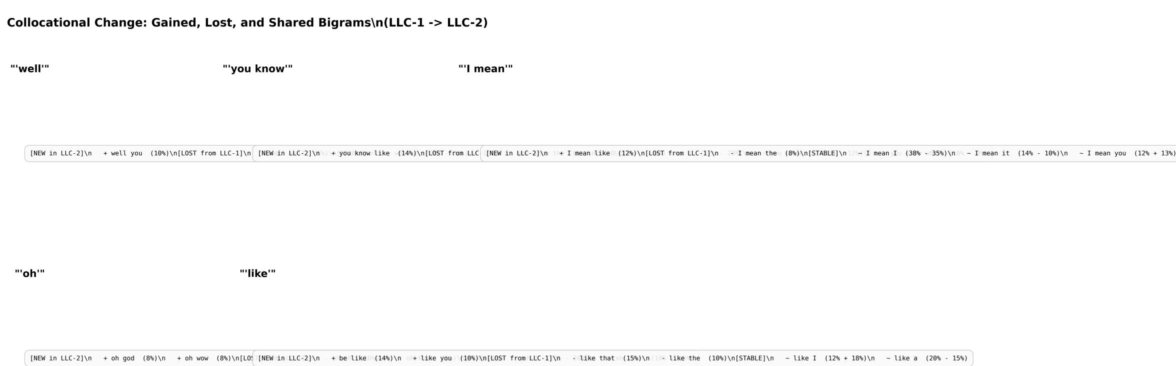

Figure 4: Collocational change summary for each marker: bigrams gained in LLC-2 (NEW), lost from LLC-1 (LOST), and shared across both periods (STABLE).

new quotative and discourse-marking functions, partially at the expense of *you know* and *I mean*. The emergence of *be like* as a major quotative construction and the collocational compounding of *like* with other discourse markers (*you know like*, *I mean like*) suggest an ongoing reorganization of the discourse marker inventory in which *like* is becoming an increasingly central procedural item. At the same time, the functional rebalancing of *well* and *oh* — fewer affective/expressive uses, more procedural/structuring uses — points to broader shifts in discourse strategies, consistent with claims about the pragmaticalization of discourse markers in Late Modern English (Brinton, 2017; Beeching, 2016).

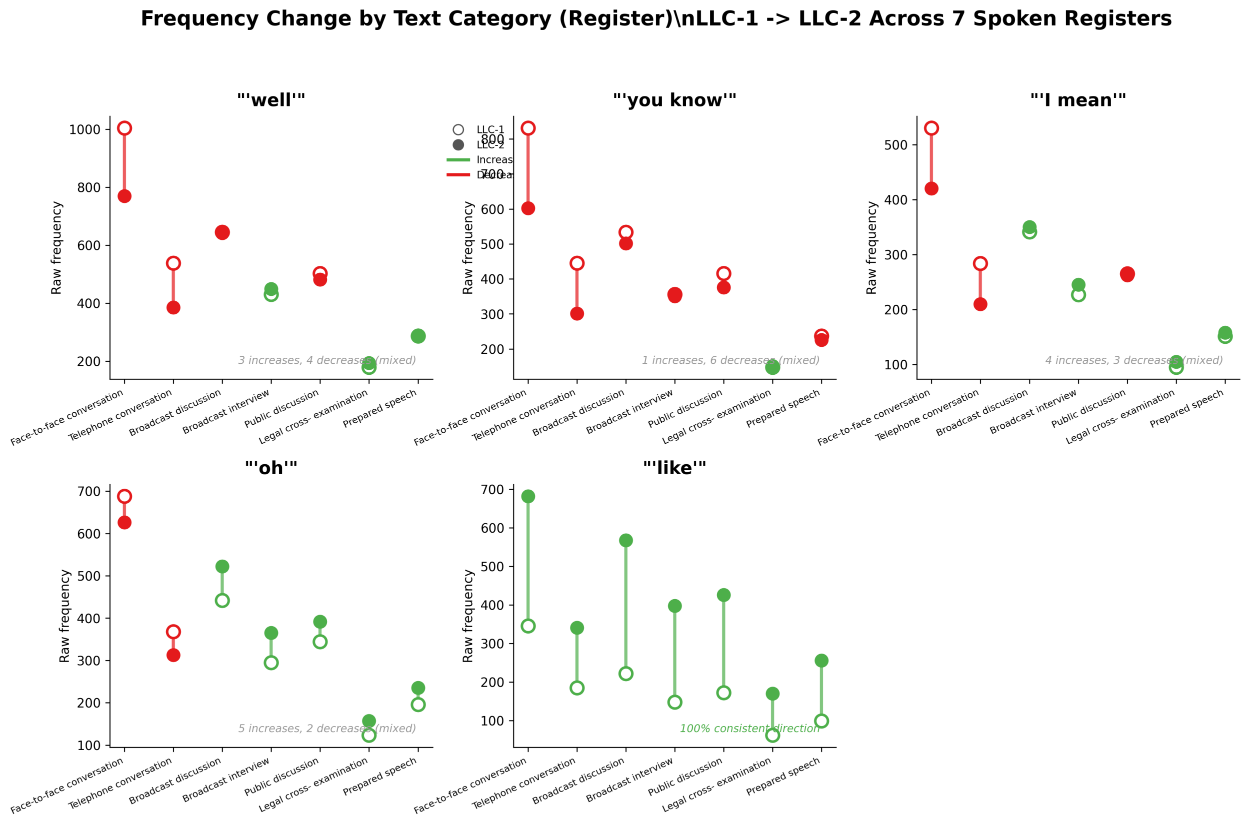

Figure 5: Text-category slope chart showing frequency per 10K words for each marker across all seven spoken registers in LLC-1 (white dots) and LLC-2 (coloured dots). Green lines = increase; red lines = decrease. Direction consistency is annotated per marker.

The Poisson regression results suggest that, overall, discourse marker usage has increased by approximately 6.5% when controlling for marker identity and register. This aggregate increase, however, masks considerable heterogeneity: *like*'s dramatic rise coexists with declines in three of the other four markers. The net trend suggests that the discourse marker system is not simply expanding or contracting but undergoing functional reorganization, with *like* assuming discourse functions that were previously more associated with *you know* and *I mean*. This pattern is consistent with usage-based accounts of grammaticalization (Bybee, 2010), in which high-frequency items undergo semantic bleaching and functional expansion, and with the documented trajectory of *like* across varieties of English (D'Arcy, 2017; Romaine and Lange, 1991).

7 Conclusion

Several limitations should be acknowledged. First, the data are published aggregate frequencies rather than token-level extractions from the full corpora; future work with direct corpus access should verify these patterns using raw annotation files. Second, the functional coding in Aijmer (2002) reflects a single annotator's analysis without systematic inter-annotator reliability assessment. Third, transcription convention differences between the two corpora may affect the comparability of specific markers, particularly for hesitations and filled pauses. Fourth, the 500K-word corpus size limits statistical power for low-frequency functions and register sub-analyses.

Despite these constraints, the matched design of LLC-1 and LLC-2 — unique in spoken corpus linguistics — provides unusually strong grounds for diachronic inference. The matched corpus architecture controls for size, register distribution, and text category structure, allowing us to attribute the observed differences to genuine language change rather than corpus design artifacts. Future research should extend this analysis to additional markers (e.g., *actually*, *basically*, *sort of*, *just*), integrate the prosodic annotations available in both corpora, and directly compare the British English patterns documented here with parallel developments in other varieties of English (Tagliamonte and D'Arcy, 2007; D'Arcy, 2017; Love et al., 2017).

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